

AFTAHİ ARDI

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Education

University of British Columbia

May 2027

Bachelor of Applied Science in Computer Engineering; CGPA: 3.70

Vancouver, BC

- Relevant Courses: Digital Systems Design, Computer Architecture, Operating Systems, Data Structures and Algorithms, Software Construction, Computer Networking, Relational Databases

Skills

Languages: Python, Java, C, C++, SystemVerilog, ARM Assembly, VHDL, SQL, JavaScript, TypeScript
Tools: Git/GitHub, Linux, Bash, SSH, Quartus, ModelSim, GitHub Actions, RTOS, Jenkins, STM32, Docker
Protocols: SPI, I2C, UART, RS485, CAN, JTAG, PCIe, PCM

Work Experience

AMD (Advanced Micro Devices)

Sept. 2025 – April 2026

Software Engineer Intern | Python, Docker, ROCm, Linux

Vancouver, BC

- Executed system-level **firmware validation** during platform bring-up, performing failure triage and regression verification across **15+ AMD ROCm** firmware releases
- Built and integrated **system-level firmware test automation** for high-speed **I/O** interfaces, enabling reliable **end-to-end** execution across firmware releases and increasing automated test coverage by **25%**
- Validated **AMD Instinct MI300X/325X/350X** systems by building scalable automation and test pipelines using **Python, Bash, Docker**, and **ROCm**, accelerating **AI** training and inference across **10+** hardware platforms
- Developed a hardware management app used by **20+ developers** daily, using **React, Python** and **MySQL**

Shift Clean Energy

Sept. 2024 – Sept. 2025

Firmware Engineer Intern | C, Python, JTAG, SPI, I2C, Jenkins

Vancouver, BC

- Designed an automated **Hardware-In-the-Loop (HIL)** testing system for a Battery Management System, improving testing efficiency and test coverage by 50% using Python-based drivers and a Pytest automation framework
- Implemented a **CI/CD pipeline** for embedded firmware releases using **Jenkins**, automating cross-compilation, static analysis, and hardware-in-the-loop test stages to cut deployment time by **50%** and eliminate manual integration errors
- Developed and optimized **RS485, SPI**, and **I2C** drivers in **C**, improving real-time performance for sensor interfacing and voltage monitoring modules, and achieving a **25% increase in data accuracy** with reduced latency
- Performed **design validation tests** on PCBs by identifying and resolving design issues to ensure reliability, using oscilloscopes to analyze signal integrity and logic analyzers to debug CAN and SPI communication protocols

Zen Maker Lab

May. 2023 – Sept. 2023

Software Engineer Intern | C++, Python, STM32

Vancouver, BC

- Instructed **20+ students** in embedded firmware development using **STM32** microcontrollers and **C++**, covering hands-on projects with **GPIO, UART**, and **SPI** peripheral interfacing
- Developed **Python** automation tools to reduce manual steps and standardize **firmware** deployment across the company's embedded hardware product lines
- Developed custom **C++** firmware for **20+** 3D printers, optimizing print mechanisms for speed and reliability

Projects

OS/161 Operating System Kernel | C, MIPS Assembly, GDB

- Designed a **32-bit virtual memory system** with dynamic allocation and paging, supporting up to **4GB** of RAM
- Implemented **system calls** for **file and process management**, bridging the kernel and user space boundary
- Built **synchronization primitives** (sleep-locks, condition variables) for kernel concurrency control

Hawkeye Autonomous Imaging System | ROS 2, Python, Embedded Linux, Docker

- Built the **central coordination service** routing **30 Hz** image data between camera, inference, and control nodes
- Designed **multi-threaded callback groups** decoupling sensor and command paths, holding **latency under 50 ms**
- Wrote a **ROS 2 sensor driver node** interfacing an **IR detector** over **I2C**, timestamping reads for inference

RISC CPU Core | SystemVerilog, Quartus, ModelSim, Computer Architecture

- Designed a **5-stage pipelined 16-bit RISC processor** with **hazard detection** in **SystemVerilog** at **1 IPC**
- Implemented a custom **RISC ISA** of arithmetic, load/store, and branch ops over a **memory-mapped interface**
- Synthesized RTL in **Quartus** and verified functional correctness through **ModelSim** simulations